NOTES ON SOLUTION TO THE MOMENTUM EQUATION WITH EXPONENTIAL ν PROFILE

GPT 5.5 WITH PROMPTING FROM TWNH

1. Problem statement I

Consider the complex horizontal velocity

$$u = u^x + iu^y, \tag{1}$$

satisfying

$$Ek\,\frac{\partial}{\partial z}\left(\nu(z)\frac{\partial u}{\partial z}\right) - if u = \mathfrak{f}, \tag{2}$$

on the vertical interval

$$-H \le z \le 0.$$

The boundary conditions are

$$Ek\,\nu(0)u_z(0) = \tau^x + i\tau^y, \tag{3}$$

$$u(-H) = 0. \tag{4}$$

Here

$$\mathfrak{f} = p_x + ip_y \tag{5}$$

is the complex pressure-gradient forcing and

$$\tau = \tau^x + i\tau^y \tag{6}$$

is the complex surface wind stress.

We assume an exponentially varying vertical viscosity

$$\nu(z) = a + \beta \exp\left[-\frac{z+H}{h}\right], \tag{7}$$

with $a > 0$, $h > 0$, and $\nu(z) > 0$ on $[-H, 0]$.

The horizontal coordinates x, y enter parametrically through $\mathfrak{f}(x, y, z)$ and $\tau(x, y)$. In the derivation below this dependence is suppressed.

Date: 7Jul26.

2. Reduced form

Divide (**??**) by Ek. Then

$$\frac{\partial}{\partial z}\left(\nu u_z\right) - \alpha u = F, \tag{8}$$

where

$$\alpha = \frac{if}{Ek}, \qquad F = \frac{\mathfrak{f}}{Ek}. \tag{9}$$

Define the differential operator

$$Lu = \partial_z(\nu u_z) - \alpha u. \tag{10}$$

The problem becomes

$$Lu = F, \tag{11}$$

with boundary conditions

$$\nu(0)u_z(0) = \frac{\tau}{Ek}, \tag{12}$$

$$u(-H) = 0. \tag{13}$$

3. Homogeneous equation

The associated homogeneous problem is

$$\partial_z(\nu u_z) - \alpha u = 0. \tag{14}$$

With

$$\nu(z) = a + \beta \exp\left[-\frac{z+H}{h}\right],$$

introduce

$$X(z) = -\frac{\beta}{a}\exp\left[-\frac{z+H}{h}\right]. \tag{15}$$

Then

$$\nu(z) = a(1 - X). \tag{16}$$

Also define

$$\lambda = h\sqrt{\frac{\alpha}{a}} = h\sqrt{\frac{if}{Ek\,a}}. \tag{17}$$

Since α is complex, λ is generally complex. A consistent branch of the square root should be chosen.

In terms of X, the homogeneous equation admits a basis

$$y_+(z) = X(z)^{\lambda}{}_2F_1\left(\lambda, \lambda+1; 2\lambda+1; X(z)\right), \tag{18}$$

$$y_-(z) = X(z)^{-\lambda}{}_2F_1\left(-\lambda, 1-\lambda; 1-2\lambda; X(z)\right). \tag{19}$$

Thus the homogeneous solution is

$$u_h(z) = C_+ y_+(z) + C_- y_-(z). \tag{20}$$

4. Boundary-adapted homogeneous solutions

We now construct two homogeneous solutions adapted to the boundary conditions.

Let $\phi(z)$ satisfy

$$L\phi = 0, \qquad \phi(-H) = 0. \tag{21}$$

A convenient choice is

$$\phi(z) = y_-(-H)y_+(z) - y_+(-H)y_-(z). \tag{22}$$

Indeed,

$$\phi(-H) = y_-(-H)y_+(-H) - y_+(-H)y_-(-H) = 0.$$

Next, let $\psi(z)$ satisfy

$$L\psi = 0, \qquad \psi_z(0) = 0. \tag{23}$$

Because the homogeneous surface-stress condition is

$$\nu(0)\psi_z(0) = 0,$$

and because $\nu(0) \neq 0$, this is equivalent to $\psi_z(0) = 0$.

A convenient choice is

$$\psi(z) = y'_-(0)y_+(z) - y'_+(0)y_-(z). \tag{24}$$

Then

$$\psi_z(0) = y'_-(0)y'_+(0) - y'_+(0)y'_-(0) = 0.$$

5. Weighted Wronskian

For two homogeneous solutions ϕ and ψ, define the weighted Wronskian

$$W = \nu(z)\left[\phi(z)\psi_z(z) - \phi_z(z)\psi(z)\right]. \tag{25}$$

This quantity is independent of z. Indeed,

$$\begin{aligned} \frac{dW}{dz} &= \frac{d}{dz}\left\{\nu\left(\phi\psi_z - \phi_z\psi\right)\right\} \\ &= \phi(\nu\psi_z)_z - \psi(\nu\phi_z)_z \\ &= \phi(\alpha\psi) - \psi(\alpha\phi) \\ &= 0. \end{aligned} \tag{26}$$

6. Green's function

Define $G(z,\zeta)$ by

$$L_z G(z,\zeta) = \delta(z-\zeta), \tag{27}$$

with homogeneous boundary conditions

$$G(-H,\zeta) = 0, \tag{28}$$

$$G_z(0,\zeta) = 0. \tag{29}$$

Then the Green's function is

$$G(z,\zeta)=\begin{cases}\dfrac{\phi(z)\psi(\zeta)}{W}, & z<\zeta,\\[2mm] \dfrac{\phi(\zeta)\psi(z)}{W}, & z>\zeta.\end{cases} \tag{30}$$

At $z=\zeta$, G is continuous:

$$G(\zeta^-,\zeta)=G(\zeta^+,\zeta), \tag{31}$$

and its derivative has jump

$$\nu(\zeta)\left[G_z(\zeta^+,\zeta)-G_z(\zeta^-,\zeta)\right]=1. \tag{32}$$

7. Treatment of the nonhomogeneous surface stress

The Green's function above satisfies homogeneous surface flux condition $G_z(0,\zeta)=0$. Therefore the pressure-gradient forcing contributes

$$u_{\rm int}(z)=\int_{-H}^{0}G(z,\zeta)F(\zeta)\,d\zeta=\frac{1}{Ek}\int_{-H}^{0}G(z,\zeta)\mathfrak{f}(\zeta)\,d\zeta. \tag{33}$$

The nonhomogeneous surface stress is included as a boundary forcing.

Let q denote the imposed reduced surface flux:

$$q=\nu(0)u_z(0)=\frac{\tau}{Ek}. \tag{34}$$

Using Green's identity for

$$L=\partial_z(\nu\partial_z)-\alpha,$$

one obtains the boundary contribution

$$u_{\rm surf}(z)=-q\,G(z,0). \tag{35}$$

Hence

$$u_{\rm surf}(z)=-\frac{\tau}{Ek}G(z,0). \tag{36}$$

8. Full solution

Combining the internal pressure forcing and the surface-stress forcing gives

$$\boxed{u(z)=\frac{1}{Ek}\int_{-H}^{0}G(z,\zeta)\mathfrak{f}(\zeta)\,d\zeta-\frac{\tau}{Ek}G(z,0).} \tag{37}$$

Restoring the horizontal dependence,

$$\boxed{u(x,y,z)=\frac{1}{Ek}\int_{-H}^{0}G(z,\zeta)\left[p_x(x,y,\zeta)+ip_y(x,y,\zeta)\right]\,d\zeta-\frac{\tau^x(x,y)+i\tau^y(x,y)}{Ek}G(z,0).} \tag{38}$$

9. Summary of definitions

The viscosity is

$$\nu(z) = a + \beta \exp\left[-\frac{z+H}{h}\right]. \tag{39}$$

The transformed variable is

$$X(z) = -\frac{\beta}{a} \exp\left[-\frac{z+H}{h}\right], \tag{40}$$

so that

$$\nu(z) = a(1-X). \tag{41}$$

The complex decay parameter is

$$\lambda = h\sqrt{\frac{if}{Ek\,a}}. \tag{42}$$

The homogeneous basis functions are

$$y_+(z) = X(z)^{\lambda}{}_2F_1\left(\lambda, \lambda+1; 2\lambda+1; X(z)\right), \tag{43}$$

$$y_-(z) = X(z)^{-\lambda}{}_2F_1\left(-\lambda, 1-\lambda; 1-2\lambda; X(z)\right). \tag{44}$$

The boundary-adapted solutions are

$$\phi(z) = y_-(-H)y_+(z) - y_+(-H)y_-(z), \tag{45}$$

$$\psi(z) = y_-'(0)y_+(z) - y_+'(0)y_-(z). \tag{46}$$

The weighted Wronskian is

$$W = \nu(z)\left[\phi(z)\psi_z(z) - \phi_z(z)\psi(z)\right], \tag{47}$$

which is independent of z.

The Green's function is

$$G(z,\zeta) = \begin{cases} \dfrac{\phi(z)\psi(\zeta)}{W}, & z < \zeta, \\ \dfrac{\phi(\zeta)\psi(z)}{W}, & z > \zeta. \end{cases} \tag{48}$$

The final solution is

$$\boxed{u(x,y,z) = \frac{1}{Ek}\int_{-H}^{0} G(z,\zeta)\left[p_x(x,y,\zeta) + ip_y(x,y,\zeta)\right]d\zeta - \frac{\tau^x(x,y) + i\tau^y(x,y)}{Ek}G(z,0).} \tag{49}$$

10. Constant-viscosity limit

If $\beta = 0$, then

$$\nu(z) = a,$$

and the equation reduces to

$$au_{zz} - \alpha u = F. \tag{50}$$

Define

$$m = \sqrt{\frac{\alpha}{a}} = \sqrt{\frac{if}{Ek\,a}}. \tag{51}$$

The homogeneous solutions are

$$u_h(z) = C_1 \cosh[m(z+H)] + C_2 \sinh[m(z+H)]\,. \tag{52}$$

A bottom-Dirichlet solution is

$$\phi(z) = \sinh[m(z+H)]\,, \tag{53}$$

and a homogeneous surface-stress solution may be chosen as

$$\psi(z) = \cosh(mz), \tag{54}$$

since

$$\psi_z(0) = 0.$$

The weighted Wronskian becomes

$$W = a\,[\phi\psi_z - \phi_z\psi]\,, \tag{55}$$

which is constant.

This gives the usual finite-depth Ekman-layer solution with bottom no-slip and imposed surface stress.

11. Remarks

- The operator
$$L = \partial_z(\nu\partial_z) - \alpha$$
is formally self-adjoint for real α, but here $\alpha = if/(Ek)$ is complex. The Green's-function construction nevertheless applies for the complex-valued linear ordinary differential equation.
- The branch of
$$\lambda = h\sqrt{\frac{if}{Ek\,a}}$$
should be chosen consistently. For $f > 0$, the principal square root gives
$$\sqrt{i} = \frac{1+i}{\sqrt{2}}.$$
- The condition $\nu(z) > 0$ requires
$$a + \beta \exp\left[-\frac{z+H}{h}\right] > 0$$
for all $z \in [-H, 0]$.
- If $\beta > 0$, then $X(z) < 0$. The hypergeometric expressions should be interpreted using a consistent branch for X^λ.
- The horizontal coordinates enter only parametrically through the pressure-gradient forcing and the wind stress.

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12. Problem statement II

We consider the complex horizontal velocity

$$u = u^x + iu^y, \tag{56}$$

satisfying the vertical boundary-value problem

$$Ek\,\frac{\partial}{\partial z}\left(\nu(z)\frac{\partial u}{\partial z}\right) - if u = \mathfrak{f}, \tag{57}$$

on

$$-H \le z \le 0.$$

The boundary conditions are

$$Ek\,\nu(0)u_z(0) = \tau^x + i\tau^y, \tag{58}$$

$$u(-H) = 0. \tag{59}$$

The complex pressure-gradient forcing is

$$\mathfrak{f} = p_x + ip_y, \tag{60}$$

and the complex surface stress is

$$\tau = \tau^x + i\tau^y. \tag{61}$$

We now take the vertical viscosity to be

$$\nu(z) = a + b\exp\left[-\frac{z+H}{h}\right] + c\exp\left(\frac{z}{h}\right), \tag{62}$$

where a, b, c, h are constants with $h > 0$. We assume

$$\nu(z) > 0 \qquad \text{for all } z \in [-H, 0]. \tag{63}$$

The horizontal variables x, y enter only parametrically through

$$\mathfrak{f}(x, y, z) \quad \text{and} \quad \tau(x, y).$$

13. Reduced form

Divide (**??**) by Ek. Define

$$\alpha = \frac{if}{Ek}, \qquad F = \frac{\mathfrak{f}}{Ek}. \tag{64}$$

Then

$$\partial_z(\nu u_z) - \alpha u = F. \tag{65}$$

Define the differential operator

$$Lu = \partial_z(\nu u_z) - \alpha u. \tag{66}$$

The boundary conditions become

$$\nu(0)u_z(0) = \frac{\tau}{Ek}, \tag{67}$$

$$u(-H) = 0. \tag{68}$$

Thus the problem is

$$Lu = F, \tag{69}$$

with a bottom Dirichlet condition and a surface Neumann-flux condition.

14. Change of variables

Introduce

$$s = \exp\Big(\frac{z}{h}\Big). \tag{70}$$

Then

$$z = h \log s,$$

and the interval $[-H, 0]$ maps to

$$s \in [s_b, s_s], \qquad s_b = e^{-H/h}, \qquad s_s = 1. \tag{71}$$

The derivatives transform as

$$\frac{d}{dz} = \frac{s}{h}\frac{d}{ds}. \tag{72}$$

Now rewrite the viscosity. Since

$$\exp\left[-\frac{z+H}{h}\right] = e^{-H/h} s^{-1}, \qquad \exp\Big(\frac{z}{h}\Big) = s,$$

we have

$$\nu(z) = a + \frac{B}{s} + cs, \qquad B := be^{-H/h}. \tag{73}$$

Therefore

$$s\nu(s) = cs^2 + as + B. \tag{74}$$

Define the quadratic polynomial

$$P(s) = cs^2 + as + B. \tag{75}$$

Then

$$\nu(s) = \frac{P(s)}{s}. \tag{76}$$

The homogeneous equation

$$\partial_z(\nu u_z) - \alpha u = 0 \tag{77}$$

becomes

$$\frac{d}{ds}\left(P(s)\frac{dU}{ds}\right) - \frac{\rho}{s}U = 0, \tag{78}$$

where

$$U(s) = u(z(s)), \qquad \rho = \alpha h^2 = \frac{if h^2}{Ek}. \tag{79}$$

Equivalently,

$$P(s)U''(s) + P'(s)U'(s) - \frac{\rho}{s}U(s) = 0. \tag{80}$$

This equation has regular singular points at

$$s = 0,$$

and at the roots of $P(s)$. It is a Fuchsian equation with four regular singular points, and therefore belongs to the general Heun class.

15. Reduction to the general Heun equation

Assume first that $c \neq 0$ and that $P(s)$ has two distinct roots r_1, r_2:

$$P(s) = c(s - r_1)(s - r_2). \tag{81}$$

Explicitly,

$$r_{1,2} = \frac{-a \pm \sqrt{a^2 - 4cB}}{2c}, \qquad B = be^{-H/h}. \tag{82}$$

Introduce the dimensionless variable

$$x = \frac{s}{r_1}, \qquad A = \frac{r_2}{r_1}. \tag{83}$$

Then the singular points are

$$x = 0, \qquad x = 1, \qquad x = A, \qquad x = \infty.$$

Writing

$$U(s) = Y(x),$$

equation (**??**) becomes

$$Y'' + \left[\frac{1}{x-1} + \frac{1}{x-A}\right] Y' - \frac{Q}{x(x-1)(x-A)} Y = 0, \tag{84}$$

where

$$Q = \frac{\rho}{cr_1} = \frac{ifh^2}{Ek\,cr_1}. \tag{85}$$

The standard general Heun equation is

$$Y'' + \left(\frac{\gamma}{x} + \frac{\delta}{x-1} + \frac{\epsilon}{x-A}\right) Y' + \frac{\alpha_H \beta_H x - q}{x(x-1)(x-A)} Y = 0, \tag{86}$$

with the Fuchsian constraint

$$\alpha_H + \beta_H + 1 = \gamma + \delta + \epsilon. \tag{87}$$

Comparing (**??**) and (**??**), one possible parameter choice is

$$\gamma = 0, \qquad \delta = 1, \qquad \epsilon = 1, \qquad \alpha_H = 0, \qquad \beta_H = 1, \qquad q = Q. \tag{88}$$

Therefore the homogeneous solutions may be represented in terms of general Heun functions.

A local solution about $x = 0$ may be written symbolically as

$$Y_1(x) = \mathrm{HeunG}\left(A, Q; 0, 1, 0, 1; x\right), \tag{89}$$

with a second linearly independent solution obtained by the standard Frobenius construction. Because here $\gamma = 0$, the second solution may involve a logarithm. In practice it is often preferable to construct two independent Heun solutions about an ordinary point in the interval rather than about the singular point $x = 0$.

For the boundary-value problem below, we only require two independent homogeneous solutions. Denote them by

$$y_1(z), \qquad y_2(z), \tag{90}$$

where each is a solution of

$$Ly = 0. \tag{91}$$

Equivalently,

$$y_j(z) = Y_j\left(\frac{e^{z/h}}{r_1}\right), \qquad j = 1, 2, \tag{92}$$

where Y_j are independent solutions of (**??**).

16. Boundary-adapted homogeneous solutions

Let $\phi(z)$ be the homogeneous solution satisfying the bottom Dirichlet condition:

$$L\phi = 0, \qquad \phi(-H) = 0. \tag{93}$$

Given two independent homogeneous solutions y_1, y_2, define

$$\phi(z) = y_2(-H)y_1(z) - y_1(-H)y_2(z). \tag{94}$$

Then

$$\phi(-H) = y_2(-H)y_1(-H) - y_1(-H)y_2(-H) = 0.$$

Next let $\psi(z)$ be the homogeneous solution satisfying the homogeneous surface-flux condition:

$$L\psi = 0, \qquad \psi_z(0) = 0. \tag{95}$$

Since $\nu(0) \neq 0$, this is equivalent to

$$\nu(0)\psi_z(0) = 0.$$

Define

$$\psi(z) = y_2'(0)y_1(z) - y_1'(0)y_2(z). \tag{96}$$

Then

$$\psi_z(0) = y_2'(0)y_1'(0) - y_1'(0)y_2'(0) = 0.$$

Here primes denote differentiation with respect to z.

17. Weighted Wronskian

For two homogeneous solutions ϕ and ψ, define the weighted Wronskian

$$W = \nu(z)\left[\phi(z)\psi_z(z) - \phi_z(z)\psi(z)\right]. \tag{97}$$

This quantity is independent of z. Indeed,

$$\begin{aligned} \frac{dW}{dz} &= \frac{d}{dz}\left\{\nu\left(\phi\psi_z - \phi_z\psi\right)\right\} \\ &= \phi(\nu\psi_z)_z - \psi(\nu\phi_z)_z \\ &= \phi(\alpha\psi) - \psi(\alpha\phi) \\ &= 0. \end{aligned} \tag{98}$$

18. Green's function

Define $G(z,\zeta)$ by

$$L_z G(z,\zeta) = \delta(z-\zeta), \tag{99}$$

with homogeneous boundary conditions

$$G(-H,\zeta) = 0, \tag{100}$$

$$G_z(0,\zeta) = 0. \tag{101}$$

Then

$$G(z,\zeta) = \begin{cases} \dfrac{\phi(z)\psi(\zeta)}{W}, & z<\zeta, \\ \dfrac{\phi(\zeta)\psi(z)}{W}, & z>\zeta. \end{cases} \tag{102}$$

The Green's function is continuous at $z=\zeta$:

$$G(\zeta^-,\zeta) = G(\zeta^+,\zeta), \tag{103}$$

and its derivative has the jump

$$\nu(\zeta)\left[G_z(\zeta^+,\zeta) - G_z(\zeta^-,\zeta)\right] = 1. \tag{104}$$

19. Solution of the forced problem

The inhomogeneous problem is

$$Lu = F = \frac{\mathfrak{f}}{Ek}, \tag{105}$$

with

$$u(-H) = 0, \tag{106}$$

$$\nu(0)u_z(0) = \frac{\tau}{Ek}. \tag{107}$$

The interior pressure-gradient forcing contributes

$$u_{\mathrm{int}}(z) = \int_{-H}^{0} G(z,\zeta)F(\zeta)\,d\zeta = \frac{1}{Ek}\int_{-H}^{0} G(z,\zeta)\mathfrak{f}(\zeta)\,d\zeta. \tag{108}$$

The nonhomogeneous surface stress contributes the boundary term

$$u_{\mathrm{surf}}(z) = -\frac{\tau}{Ek}G(z,0). \tag{109}$$

Therefore the full solution is

$$\boxed{u(z) = \frac{1}{Ek}\int_{-H}^{0} G(z,\zeta)\mathfrak{f}(\zeta)\,d\zeta - \frac{\tau}{Ek}G(z,0).} \tag{110}$$

Restoring the horizontal dependence,

$$\boxed{u(x,y,z) = \frac{1}{Ek}\int_{-H}^{0} G(z,\zeta)\left[p_x(x,y,\zeta) + ip_y(x,y,\zeta)\right]\,d\zeta - \frac{\tau^x(x,y) + i\tau^y(x,y)}{Ek}G(z,0).} \tag{111}$$

20. Summary

For

$$\nu(z) = a + b\exp\left[-\frac{z+H}{h}\right] + c\exp\left(\frac{z}{h}\right), \tag{112}$$

define

$$s = e^{z/h}, \qquad B = be^{-H/h}, \tag{113}$$

and

$$P(s) = cs^2 + as + B. \tag{114}$$

The homogeneous equation becomes

$$\frac{d}{ds}\left(P(s)\frac{dU}{ds}\right) - \frac{\rho}{s}U = 0, \qquad \rho = \frac{ifh^2}{Ek}. \tag{115}$$

If $c \neq 0$ and $P(s) = c(s-r_1)(s-r_2)$, define

$$x = \frac{s}{r_1}, \qquad A = \frac{r_2}{r_1}, \qquad Q = \frac{\rho}{cr_1}. \tag{116}$$

The homogeneous equation becomes

$$Y'' + \left[\frac{1}{x-1} + \frac{1}{x-A}\right]Y' - \frac{Q}{x(x-1)(x-A)}Y = 0, \tag{117}$$

which is a general Heun equation with parameters

$$\gamma = 0, \qquad \delta = 1, \qquad \epsilon = 1, \qquad \alpha_H = 0, \qquad \beta_H = 1, \qquad q = Q. \tag{118}$$

Let y_1, y_2 be any two independent homogeneous solutions. Define

$$\phi(z) = y_2(-H)y_1(z) - y_1(-H)y_2(z), \tag{119}$$

$$\psi(z) = y_2'(0)y_1(z) - y_1'(0)y_2(z). \tag{120}$$

Then

$$W = \nu(z)\left[\phi(z)\psi_z(z) - \phi_z(z)\psi(z)\right] \tag{121}$$

is constant, and

$$G(z,\zeta) = \begin{cases} \dfrac{\phi(z)\psi(\zeta)}{W}, & z < \zeta, \\ \dfrac{\phi(\zeta)\psi(z)}{W}, & z > \zeta. \end{cases} \tag{122}$$

The velocity is

$$\boxed{u(x,y,z) = \frac{1}{Ek}\int_{-H}^{0} G(z,\zeta)\left[p_x(x,y,\zeta) + ip_y(x,y,\zeta)\right]\,d\zeta - \frac{\tau^x(x,y) + i\tau^y(x,y)}{Ek}G(z,0).} \tag{123}$$

21. Special cases

21.1. **Case $c = 0$.** If $c = 0$, then

$$\nu(z) = a + b \exp\left[-\frac{z+H}{h}\right].$$

This is the single-exponential case. The homogeneous equation reduces to a Gauss hypergeometric equation. With

$$X(z) = -\frac{b}{a} \exp\left[-\frac{z+H}{h}\right],$$

and

$$\lambda = h\sqrt{\frac{if}{Ek\,a}},$$

a homogeneous basis is

$$y_+(z) = X(z)^{\lambda}\,{}_2F_1\left(\lambda, \lambda+1; 2\lambda+1; X(z)\right), \tag{124}$$

$$y_-(z) = X(z)^{-\lambda}\,{}_2F_1\left(-\lambda, 1-\lambda; 1-2\lambda; X(z)\right). \tag{125}$$

21.2. **Case $b = 0$.** If $b = 0$, then

$$\nu(z) = a + c \exp\left(\frac{z}{h}\right).$$

This is also reducible to a Gauss hypergeometric equation by using

$$X(z) = -\frac{c}{a} e^{z/h}.$$

21.3. **Constant-viscosity limit.** If $b = c = 0$, then

$$\nu(z) = a,$$

and the equation becomes

$$a u_{zz} - \alpha u = F. \tag{126}$$

Define

$$m = \sqrt{\frac{\alpha}{a}} = \sqrt{\frac{if}{Ek\,a}}. \tag{127}$$

The homogeneous solution is

$$u_h(z) = C_1 \cosh[m(z+H)] + C_2 \sinh[m(z+H)]. \tag{128}$$

A bottom-Dirichlet solution is

$$\phi(z) = \sinh[m(z+H)], \tag{129}$$

and a homogeneous surface-flux solution is

$$\psi(z) = \cosh(mz), \tag{130}$$

since

$$\psi_z(0) = 0.$$

This gives the usual finite-depth Ekman solution with bottom no-slip and imposed surface stress.

22. Remarks

- The two-exponential viscosity profile
$$\nu(z) = a + be^{-(z+H)/h} + ce^{z/h}$$
leads to a general Heun equation, whereas the one-exponential profile leads to a Gauss hypergeometric equation.
- The Green's-function representation does not depend on the particular normalization of y_1, y_2. Any two independent homogeneous solutions may be used.
- Since
$$\alpha = \frac{if}{Ek},$$
the homogeneous solutions are generally complex-valued.
- The branch choices in the Heun functions must be made consistently if the singularities lie near the physical interval.
- The physical interval in the s-variable is
$$e^{-H/h} \le s \le 1.$$
The solution is regular provided $P(s) \neq 0$ on this interval, which is equivalent to $\nu(z) > 0$.
- If a root of $P(s)$ lies on the physical interval, then the viscosity vanishes and the boundary-value problem becomes singular.

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